MIL-D-18665B(AS)
21 March 1969

SUPERSEDING
MIL-D-18665A (Wep)
23 February 1961

# MILITARY SPECIFICATION

# DETONATOR MARK 59 MOD 0

This specification has been approved by the Naval Air Systems Command, Department of the Navy.

## 1. SCOPE

1.1 This specification governs the manufacture, assembly, loading, and preparation for delivery of the Detonator Mark 59 Mod 0 and the methods of inspection and tests upon which acceptance will be based. Detonator Mark 59 Mod 0 is flash initiated.

## 2. APPLICABLE DOCUMENTS

2.1 The following specifications, standards, drawings, and publications, of the issue in effect on date of invitation for bids, form a part of this specification:

SPECIFICATIONS

MILITARY

| | |
|---|---|
| JAN-T-339 | Tetryl (trinitrophenylmethyl-nitramine) |
| MIL-L-3055 | Lead Azide |

STANDARDS

| | |
|---|---|
| MIL-STD-105 | Sampling Procedures and Tables for Inspection by Attributes |
| MIL-STD-129 | Marking for Shipment and Storage |
| MIL-STD-1314(NAVY) | Safety Precautions for Explosive Loaded Items |
| MIL-STD-331 | Fuze and Fuze Components Environmental and Performance Tests for |



Source: https://assist.dla.mil -- Downloaded: 2026-10-18T11:48Z
Check the source to verify that this is the current version before use.

DRAWINGS

NAVAL AIR SYSTEMS COMMAND

| | |
|---|---|
| LD 174855 | Detonator Mk 59 Mod 0 |
| LD 174879 | Test Set Adapter Assembly for Detonator Mk 59 Mod 0 |

NAVAL ORDNANCE SYSTEMS COMMAND

| | |
|---|---|
| LD 166538 | Test Set Mk 136 Mod 0 |
| LD 275959 | Detonator Container Mk 6 Mod 0 (Inner) |
| LD 275960 | Detonator Container Mk 8 Mod 0 (Outer) |
| LD 290814 | Primer Mk 125 Mod 1 |

PUBLICATIONS

NAVAL ORDNANCE SYSTEMS COMMAND

| | |
|---|---|
| OD 5823 | Sensitivity Test of Primers and Detonators using Test Set Mk 135 Mod 0 (Primer) and Test Set Mk 136 Mod 0 (Detonator) |

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the Procuring Activity or as directed by the Contracting Officer.)

2.2 Other publications. - The following documents form a part of this specification. Unless otherwise indicated the issue in effect on date of invitation for bids shall apply.

CODE OF FEDERAL REGULATIONS

| | |
|---|---|
| Title 49 - Transportation Parts 171-190 | Regulations for Transportation of Explosives and Other Dangerous Articles |

(The Code of Federal Regulations are available from the Superintendent of Documents, Government Printing Office, Washington, D. C. 20402.)



Source: https://assist.dla.mil -- Downloaded: 2026-10-18T11:48Z
Check the source to verify that this is the current version before use.

## 3. REQUIREMENTS

### 3.1 General

3.1.1 Construction. - Detonators covered by this specification shall be manufactured in accordance with Naval Air Systems Command - LD No. 174855 and all documents listed thereon.

3.1.2 Preproduction sample. - Unless otherwise specified in the contract or order, a preproduction sample of 315 Detonators Mk 59 Mod 0 shall be manufactured using the methods and procedures and at the same location as proposed for full production. This sample shall be tested as specified in 4.2.2 herein and is for the purpose of determining that the contractor, prior to starting production, is capable of producing units which comply with the technical requirements of the contract including the drawings and this specification.

3.1.3 Inspection requirements. - The Detonators Mk 59 Mod 0 shall be inspected to determine conformance with the requirements of this specification, LD 174855 and all drawings and documents listed thereon. Inspection shall be in accordance with 4.4, 4.5 and Table I.

3.1.4 Data. - No data is required by applicable documents referenced in Section 2, unless specified in the contract or order.

### 3.2 Explosive materials.

3.2.1 Lead azide. - The lead azide shall conform to the requirements of specification MIL-L-3055. The moisture content of the lead azide at the time of loading shall not exceed 0.30 percent as determined by the procedure of 4.5.1.

3.2.2 Tetryl. - The tetryl shall conform to the requirements of specification JAN-T-339, Grade I, Class B. The volatiles and moisture content of the tetryl at the time of loading shall not exceed 0.10 percent when tested as specified in 4.5.2.

3.3 Detonator functioning. - The Detonator Mk 59 Mod 0 shall function when exposed to the flash of the Primer Mk 125 Mod 1 over a 1.00 inch air gap, using the Test Set Adapter Assembly for Detonator Mk 59 Mod 0, LD. No. 174879 in conjunction with Test Set Mk 136 Mod 0. Testing and acceptance shall be in accordance with 4.5.3.



Source: https://assist.dla.mil -- Downloaded: 2026-10-18T11:48Z
Check the source to verify that this is the current version before use.

3.4 Detonator output. - The explosion of the Detonator Mk 59 Mod 0 in the Test Set Mk 136 Mod 0 using the Test Set Adapter Assembly for Detonator Mk 59 Mod 0, LD No. 174879 shall produce an indentation in the steel block having a depth equal to or greater than .0165 inch. Testing and acceptance shall be in accordance with 4.5.4.

## 4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. - Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Sampling plans and procedures.

4.2.1 Inspection lot. - Lot definition, formation and size shall be in accordance with MIL-STD-105.

4.2.2 Preproduction sample. - The preproduction sample of 315 Detonators, Mk 59 Mod 0 shall be inspected as specified in 4.5, Figure 2, and Table I for the purpose of determining compliance with the requirements of the contract, specifications, and drawings. Further production of the item by the contractor prior to the approval of the contracting activity shall be at the contractor's risk.

4.2.2.1 Preproduction sample for a subsequent contract. - The necessity for a preproduction sample will be determined by the procuring agency when production under a new contract by the same contractor at the same location follows the manufacture of satisfactory material as covered by this specification.

4.3 Acceptance test equipment and conditions. - The following items of test equipment are required to perform the acceptance tests of this specification as shown in Figure 3.

| | |
|---|---|
| Test Set Mk 136 Mod 0 | LD 166538 |
| Test Set Adapter Assembly for Detonator Mk 59 Mod 0 | LD 174879 |



Source: https://assist.dla.mil -- Downloaded: 2026-10-18T11:48Z
Check the source to verify that this is the current version before use.

4.3.1 Test conditions. - Unless otherwise specified, the detonators shall be subjected to acceptance tests under the following conditions:

4.3.1.1 Temperature: Room ambient 65 degrees F. to 95 degrees F.

4.3.1.2 Humidity: Room ambient to 95 percent relative, maximum.

4.4 Quality Conformance Inspection. - The detonators shall be inspected to determine conformance with this specification and LD 174855. Inspection shall be conducted as specified in Table I and Figure 3. Acceptance and rejection shall be in accordance with MIL-STD-105 and the stated AQL unless otherwise required by special sampling procedure.

4.5 Test procedures.

4.5.1 Moisture content test for lead azide. - Conduct at least one test for each lot of loaded detonators and such additional tests warranted by variations in the process and materials. Collect an approximate two gram sample in a tared, stoppered container at the site of and immediately prior to consolidating the charge in the detonator. Weigh the container; open and expose the sample for drying at 50 $\pm$ 2 degrees C and at least 28 inches (Hg) vacuum until constant weight is reached. Cool the container in a desiccator; stopper and weigh. All weighings shall be accurate to $\pm$ 0.5 milligram. The presence of moisture in excess of 0.30 percent shall cause suspension of the charge loading until permanent corrective measures have been completed. Detonators which have been loaded with the defective charges shall be rejected.

4.5.2 Volatiles and moisture content test for tetryl. - Conduct at least one test for each lot of loaded detonators and such additional tests warranted by variations in the process and materials. Test procedure shall be in accordance with that given in specification JAN-T-339. The presence of volatiles and moisture in excess of 0.10 percent shall cause suspension of the charge loading until permanent corrective measures have been completed. Detonators which have been loaded with the defective charges shall be rejected.



Source: https://assist.dla.mil -- Downloaded: 2026-10-18T11:48Z
Check the source to verify that this is the current version before use.

4.5.3 Detonator functioning test. - A sample of the Detonators Mk 59 Mod 0 shall be selected and tested from each lot in accordance with the Multiple Sampling Plan, Inspection Level II, AQL 0.25 percent defective as given in MIL-STD-105. The initiation of the detonator shall be accomplished by the use of Test Set Mk 136 Mod 0, LD No. 166538, in conjunction with Test Set Adapter Assembly for Detonator Mk 59 Mod 0, LD. 174879 and Test Ball, Drawing No. 553491 piece 3, firing a Primer Mk 125 Mod 1. The two ounce test ball shall be dropped from a height of 18 inches to fire the primer. The procedure for operating the test set is described in NAVORD OD 5823. Detonators which fail to explode on the first trial shall be classed as defective unless such failure is definitely attributable to a failure of the Primer Mk 125 Mod 1 to fire. In the latter case, that particular detonator test shall be disregarded. Acceptance of the lot shall be in accordance with the sampling plan given above. The steel blocks of the Detonators Mk 59 Mod 0 shall be retained and used to fill the sample requirements of 4.5.4.

4.5.4 Detonator output test. - A sample of Detonators Mk 59 Mod 0 shall be selected and tested in accordance with the Multiple Sampling Plan, Inspection Level I, AQL 1.0 percent defective as given in MIL-STD-105. The steel blocks retained from the functioning test of 4.5.3 shall be used to fill sample requirements. When necessary continue the test firing of 4.5.3 to provide the required number of steel blocks. If an excess of blocks are available from 4.5.3 random selection of the sample shall be practiced. A detonator which produces an indentation in the steel block of less than 0.0165 inch in depth as determined in accordance with 4.5.4.1 shall be classed as defective. The lot shall be rejected if the sample fails to meet the conditions of the above sampling plan.

4.5.4.1 Indentation depth measurement procedure. - After firing the detonator, remove the Steel Block, Drawing 2280950 for depth of indentation measurements. Before measurements remove any foreign material, such as deposits, from the indentation. Measurement of the depth of indentation shall be made with dial indicator capable of measuring 0.0001 inch units. The point of the dial indicator probe shall have a 30° ± 1° included angle and the end of the point shall have a radius of 0.025 ± 0.002 inch. Zero the indicator with the point of the probe in the deepest part of the indentation and take the readings at four points near the cut face edges of the steel block. These points shall be approximately 0.125 inch in from the center of each of the four cut face edges of the indented surface. The average of these four readings shall be recorded as the indentation depth of the individual steel block.



Source: https://assist.dla.mil -- Downloaded: 2026-10-18T11:48Z
Check the source to verify that this is the current version before use.

## 5. PREPARATION FOR DELIVERY

5.1 Packaging.

5.1.1 Level A.

5.1.1.1 Unit packaging. - One thousand (1000) detonators shall be packaged in accordance with LD 275959.

5.1.2 Level B. - Not applicable.

5.1.3 Level C. - Not applicable.

5.2 Packing.

5.2.1 Level A.

5.2.1.1 Exterior container. - Four thousand (4000) detonators packaged as specified in 5.1.1.1 shall be packed in accordance with LD 275960.

5.2.2 Level B. - Not applicable.

5.2.3 Level C. - Not applicable.

5.3 Marking.

5.3.1 Special markings. - All special or precautionary marking, shall be in accordance with MIL-STD-129 and Code of Federal Regulations 49 CFR 71-78.

5.3.2 Normal markings. - In addition to any special markings required by the contract or order, all containers shall be marked in accordance with Military Standard MIL-STD-129.

## 6. NOTES

6.1 Intended use. - The Detonator Mk 59 Mod 0 is intended for use in explosive trains of fuzes and firing devices.

6.2 Ordering data. - Procurement documents should specify:

(a) The specific title, number and date of this specification and exceptions to this specification, applicable drawings and other documents.



Source: https://assist.dla.mil -- Downloaded: 2026-10-18T11:48Z
Check the source to verify that this is the current version before use.

(b) Completeness and availability of test records and data cards desired.

(c) Authorization for the procurement of tetryl and lead azide as required in 3.2.1 and 3.2.2 and the Primer Mk 125 Mod 1 as required in 4.5.3 and 4.5.4.

6.3 Explosives safety precautions. - Explosive safety precautions shall be in accordance with MIL-STD-1314 (NAVY), Safety Precautions for Explosive Loaded Items.

6.4 Data cards. - Data cards when required by the contract or order shall be five inches by eight inches, printed as shown in Figure 1, shall be filled out for each lot of primers. Two of these cards shall accompany the shipping ticket; these shall be forwarded to the Procuring Agency and five shall be furnished the Inspector.

6.5 Advisory detonator output test information. - MIL-STD-331, Test 301 - Detonator Output Measurement by the Steel Disc Test provides advisory reference information on the detonator output test.



Source: https://assist.dla.mil -- Downloaded: 2026-10-18T11:48Z
Check the source to verify that this is the current version before use.

FIGURE 1

Detonator Data Card

Mark No._____Mod No._____ Output Test Record Data No._____

Dwg. No._____Rev._____ Lot No._____

Loaded by_____ Spec.No._____Rev._____

Contract No._____ Date (Mo. & Yr.)_____

No. in Box_____ Req. No._____

Steel Block Indentation Depth Av._____Min._____Max._____

| MATERIAL | MANUFACTURER | LOT OR BATCH No. | Date of Loading (Mo. & Yr.) |
|---|---|---|---|
| DETONATOR CUP | | | |
| LEAD AZIDE CLOSURE DISC | | | |
| TETRYL CLOSURE DISC | | | |
| LEAD AZIDE | | | |
| TETRYL | | | |
| DETONATOR | | | |

Remarks: (These are to include any changes which, in the opinion of the Inspector, would destroy the uniformity of the lot. They may be noted on the back of the card.)



Source: https://assist.dla.mil -- Downloaded: 2026-10-18T11:48Z
Check the source to verify that this is the current version before use.

PREPRODUCTION TESTS CONDUCTED BY CONTRACTOR

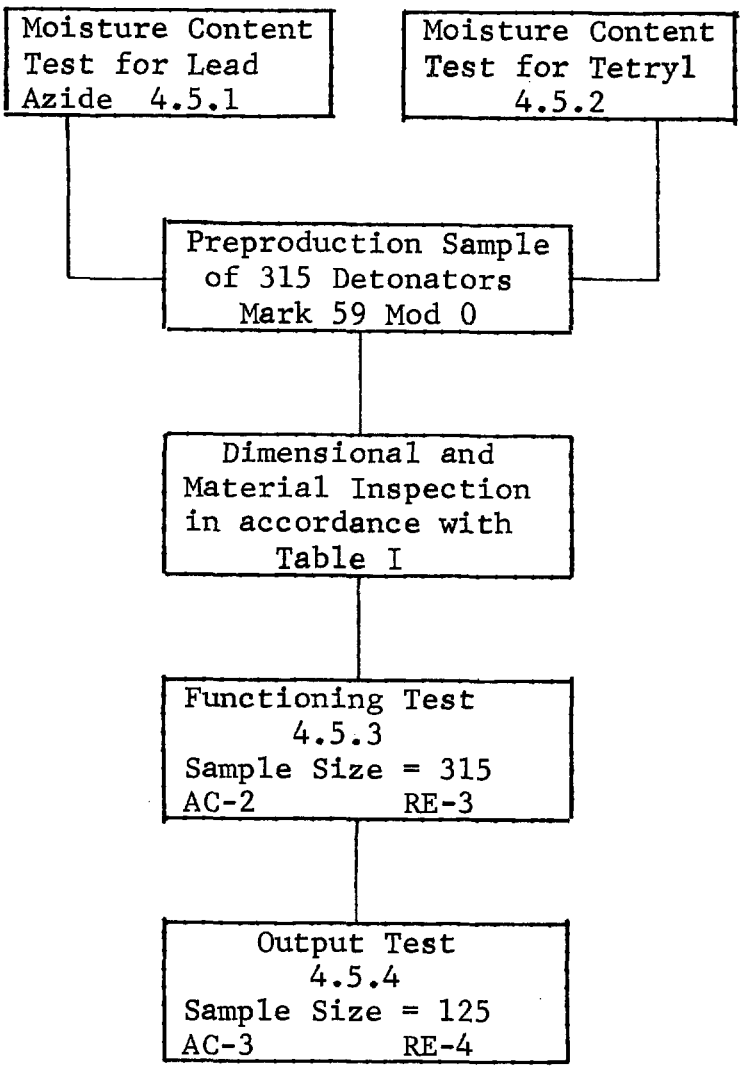



FIGURE 2



Source: https://assist.dla.mil -- Downloaded: 2026-10-18T11:48Z
Check the source to verify that this is the current version before use.

QUALITY CONFORMANCE TESTS CONDUCTED BY CONTRACTOR

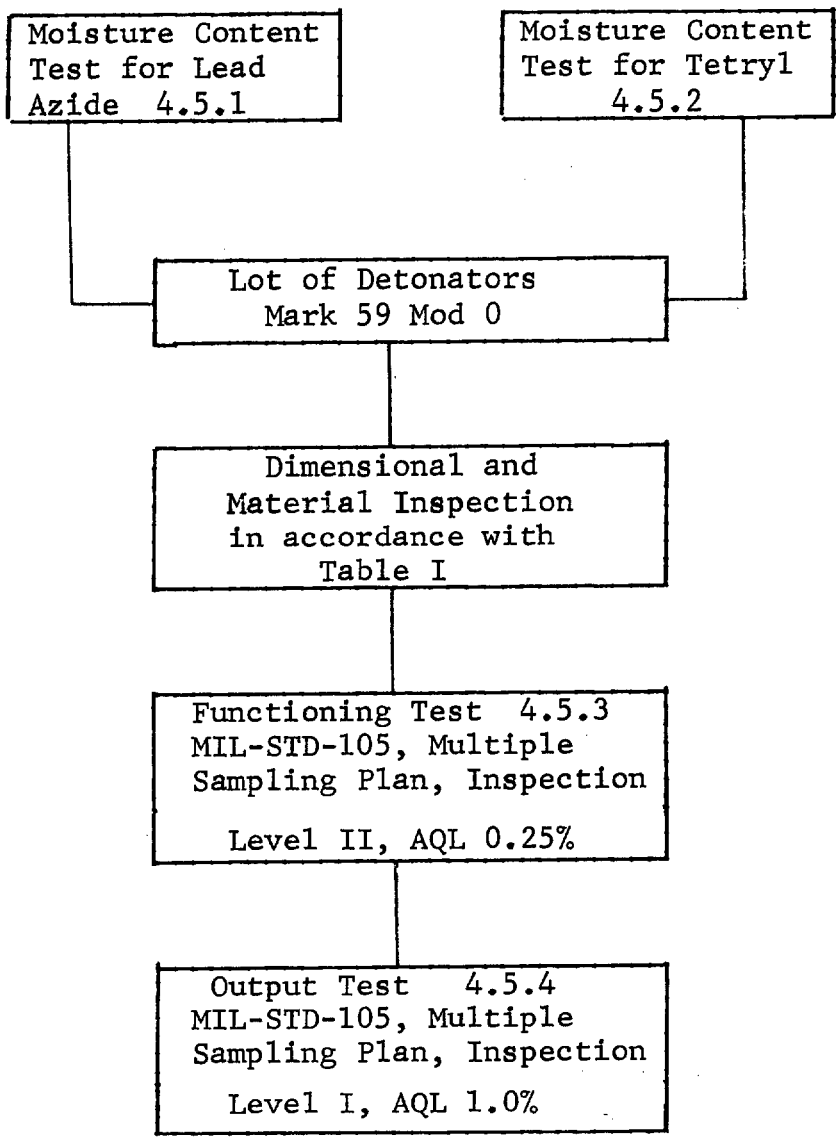



FIGURE 3



Source: https://assist.dla.mil -- Downloaded: 2026-10-18T11:48Z
Check the source to verify that this is the current version before use.

TABLE I

CLASSIFICATION OF DEFECTS
FOR DETONATOR MARK 59 MOD 0
(BUWEPS DWG 652220)

NOTES:

A. All dimensions are in inches unless otherwise specified.

B. Requirements preceded by an asterisk (*) shall be verified prior to or during loading of detonator.

C. Paragraph references apply to this specification.

D. Reference LD 174855.

E. Requirements followed by (S) shall be verified in accordance with the applicable specification. When sampling plans are not given, requirements shall be verified to the satifaction of the Government Inspector.

| | REQUIREMENT | REFERENCE |
|---|---|---|
| CRITICAL (100% Inspection) | | |
| 1. | Assembly shall be free of splits, tears and cracks | 652220 |
| 2. | Exterior surfaces of detonator must be free of exposed explosives and explosive contamination (macroscopic) | 652220 |
| 3. (S) | Preparation for delivery shall be in accordance with specification | Para 5 |
| MAJOR | | |
| Inspect requirements below using an AQL of 0.25% defective unless otherwise specified. | | |
| *101. | Explosives shall be loaded in the sequence shown in the drawing | 652220 |
| 102. | Detonators shall pass functioning test | Para 4.5.3 |



Source: https://assist.dla.mil -- Downloaded: 2026-10-18T11:48Z
Check the source to verify that this is the current version before use.

| | REQUIREMENT | REFERENCE |
|---|---|---|
| | Inspect requirements below using an AQL of 1.0% defective unless otherwise specified. | |
| 151.(S) | Detonators shall pass output test | Para 4.5.4 |
| *152. | .0020 ± .0005 MAX. | 652222 |
| *153. | .010 ± .001 MAX. | 652223 |
| *154.(S) | Lead azide (.116 ± .002 g. approximate), MIL-L-3055, consolidated in two equal increments at 10,000 ± 300 psi | 652220 |
| *155.(S) | Tetryl (.060 g. approximate) JAN-T-339, Grade I, Class B, consolidated in two equal increments. Alt. Tetryl may be pelletized into one increment of .060 ± .003 g. and reconsolidated at a pressure of 10,000 ± 300 psi. The length of the preformed tetryl pellet shall not be less than 0.192. Use of graphite, JAN-G-155 not to exceed 2% is permitted | 652220 |
| *156.(S) | Moisture content in lead azide must not exceed .30% | Para 4.5.1 |
| *157. | .161 DIA MAX. | 652220 |
| *158. | .380 + .000 - .010 MAX. | 652220 |
| 159. | Detonator ends must be sealed with varnish in accordance with drawing | 652220 |
| *160.(S) | Volatiles and moisture content in tetryl must not exceed .10% | Para 4.5.2 |
| *161.(S) | Aluminum Sheet, QQ-A-561 Condition A | 652221 and 652223 |
| | MINOR - (AQL 2.5%) | |
| *201. | .125 + .000 - .003 DIA. MAX. | 652222 |
| *202. | .125 + .000 - .003 DIA. MAX. | 652223 |
| *203. | .410 ± .005 | 652221 |
| *204. | .129 + .000 - .002 DIA. | 652221 |
| *205. | .089 + .000 - .004 DIA. | 652221 |
| *206. | .1595 + .000 - .002 DIA. | 652221 |
| *207. | .089 DIA. must be concentric with .1595 DIA. within .002 TIR | 652221 |
| *208. | Detonator cup must be free of oil and grease | 652221 |
| *209. | 45° ± 5° Chamfer | 652221 |
| *210.(S) | Aluminum sheet, QQ-A-561 Condition A | 652222 |





Source: https://assist.dla.mil - Downloaded: 2026-10-18T11:48Z
Check the source to verify that this is the current version before use.

# SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION

MIL-D-18665B(AS) DETONATOR MARK 59 MOD 0

| ORGANIZATION *(Of submitter)* | CITY AND STATE |
|---|---|
| | |

| CONTRACT NO. | QUANTITY OF ITEMS PROCURED | DOLLAR AMOUNT |
|---|---|---|
| | | $ |

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

| SUBMITTED BY *(Printed or typed name and activity)* | DATE |
|---|---|
| | |

DD FORM 1 APR 63 1426

C-8279

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T11:48Z
Check the source to verify that this is the current version before use.

FOLD

DEPARTMENT OF THE NAVY

Naval Air Systems Command
Washington, D.C. 20360

POSTAGE AND FEES PAID
NAVY DEPARTMENT

OFFICIAL BUSINESS

NAVAL AIR SYSTEMS COMMAND (AIR-52021)
DEPARTMENT OF THE NAVY
WASHINGTON, D. C. 20360

FOLD

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T11:48Z
Check the source to verify that this is the current version before use.